Supplement Material

Scalable Approximate Algorithm for Dynamic Densest Subhypergraphs with Solution-Guided Maintenance

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A TRANSPORT BOUNDS AND SUPPORTING DERIVATIONS

We collect the proofs of the search-work and transport bounds, using the notation of Sections 5.1 and 5.2. We also give the supporting update-charge inequalities and the degree-tail bound.

A.1 Proof of the Local Search-Work Bound

Proof of Lemma 5.3. Consider one search and set $H = \{v : b\deg(v) \geq t\}$, so $|H| \leq h$. Let P count inspected positive incidences, E distinct expanded hyperedges, and J endpoint tests. All expanded vertices have load at least t and belong to H. Each positive incidence carries at least one integer unit, so their total number is bounded by the sum of loads over H. Also, every expanded edge consumes an inspected incidence. Therefore $P \leq hQ$ and $E \leq P$.

A fully expanded edge contains only vertices of load at least t: otherwise the search would have stopped at an eligible destination. Such an edge lies in H and has at most $\min\{r, h\}$ endpoints. The last edge of a successful search may leave H, but scanning it stops at the first lower-load endpoint. If σ indicates success, these observations give

$$J \leq \min\{r, h\}E + \sigma,$$
$$P + J \leq (1 + \min\{r, h\})hQ + \sigma.$$

With K successful searches, summing all $K + F$ searches gives

$$S_{\text{scan}} \leq (1 + \min\{r, h\})hQ(K + F) + K.$$

Each successful transfer moves a positive integer amount, so $K \leq A$, which proves Equation (8). □

A.2 Proof of the Transport Bound

A.2.1 Potential Decrease under a Transfer. Set $\delta = \epsilon/\alpha$, so that $c = \lfloor s \rfloor$, $t = \lfloor \beta s \rfloor$, and $1 - \beta = \delta/2$. We use the potential

$$\Phi_s(q) = \frac{1}{s}\sum_v (q_v - \beta s + 1)_+^2, \qquad (x)_+ = \max\{x, 0\},$$

and assign potential zero to an empty state. It charges only loads above $\beta s - 1$, quadratically in their excess. An eligible destination has $q_w < t$ and contributes zero before the transfer. The shift by one handles integer rounding.

Lemma A.1 (Potential decrease of a reserve transfer). *At a fixed scale s, a successful reserve path of amount a satisfies*

$$\Phi_s(q^{\text{before}}) - \Phi_s(q^{\text{after}}) \geq \delta a. \tag{A.1}$$

Proof of Lemma A.1. Write $a_s = \beta s - 1$. Only the root z and destination w change their loads. An eligible integer destination satisfies $q_w \leq t - 1 \leq a_s$, so its initial contribution is zero and its final unscaled contribution is at most a^2. By Equation (4), the root ends at $q_z - a \geq c$. Since $c - a_s \geq (s-1) - (\beta s - 1) = (1-\beta)s > 0$, its decrease before division by s is

$$\begin{aligned}(q_z - a_s)^2 - (q_z - a - a_s)^2 &= 2a(q_z - a - a_s) + a^2 \\ &\geq 2a(c - a_s) + a^2.\end{aligned}$$

Subtracting the destination's increase and dividing by s leaves at least $2a(1-\beta) = \delta a$. □

Each transported unit releases at least δ units of potential, charging repairs to potential introduced by updates. Convex load potentials have precedent [2]; our scale tracks the maintained solution and may decrease after deletion.

A.2.2 Properties of the Shifted Potential. Write $a_s = \beta s - 1$. From $bL \geq 2/\epsilon$, we obtain

$$s \geq \frac{2(1+\epsilon)}{\epsilon} = \frac{2}{\delta}, \qquad a_s = (1 + \epsilon/2)bL - 1 \geq \frac{2}{\epsilon}.$$

In particular, $a_s \geq 0$ and $1/s \leq \delta/2$. For $q > a_s$, the derivative of $(q - a_s)^2/q$ is $1 - a_s^2/q^2 \geq 0$. For $q \leq a_s$, its positive-part version is zero. Consequently, for every stable load $0 < q \leq c \leq s$,

$$\frac{(q - a_s)_+^2}{sq} \leq \frac{(s - a_s)^2}{s^2} = (1 - \beta + 1/s)^2.$$

Summing over vertices and using $\sum_v q_v = bm$ gives

$$\Phi_s(q) \leq (1 - \beta + 1/s)^2 bm \leq \delta^2 bm. \tag{A.2}$$

The last inequality uses $1 - \beta = \delta/2$. We also need monotonicity in the pilot-solution scale. For a fixed load q, an active summand satisfies

$$\frac{\partial}{\partial s}\frac{(q + 1 - \beta s)^2}{s} = -\frac{(q+1)^2}{s^2} + \beta^2 < 0. \tag{A.3}$$

An inactive summand is zero, and the two pieces meet continuously with zero derivative. For a fixed allocation, increasing s cannot increase potential, even if a coordinate becomes inactive.

A.2.3 Single-Update Charges.

Lemma A.2 (Insertion charge). *A complete insertion transports at most b units and increases the stable potential by at most $2\delta b$.*

This bounds both effects of an insertion: the repair performed now and the potential it can leave for later repairs. Since an inserted edge adds exactly b units, the first bound is independent of the number of other edges in the graph.

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Proof of Lemma A.2. Let s_i and s_f be the initial and final scales. An insertion cannot decrease pilot-solution density, so $s_f \geq s_i$ when the initial state is nonempty. Fix s_f throughout this argument and define

$$h(q) = \frac{(q - \beta s_f + 1)_+^2}{s_f}.$$

Extend it above s_f by its tangent:

$$g(q) = \begin{cases} h(q), & q \leq s_f, \\ h(s_f) + h'(s_f)(q - s_f), & q > s_f. \end{cases}$$

The tangent extension g is convex and nondecreasing. Its derivative increases up to $h'(s_f)$ and is constant thereafter, so it is globally Lipschitz with constant

$$\kappa = 2(1 - \beta + 1/s_f) \leq 2\delta.$$

The new edge adds nonnegative amounts summing to b. Their total increase in $\sum_v g(q_v)$ is therefore at most κb. Every subsequent transfer obeys $q_z - a \geq c \geq q_w + a$. For a convex function, transferring a from the larger coordinate towards the smaller under this condition gives $g(q_z - a) + g(q_w + a) \leq g(q_z) + g(q_w)$. Thus transfers can only decrease the sum, and changes of the pilot solution leave it unchanged because the allocation is fixed.

Initial and final loads are at most s_f, where $g = h$. Writing these vectors as q^i, q^f, respectively, yields

$$\Phi_{s_f}(q^f) \leq \Phi_{s_f}(q^i) + 2\delta b \leq \Phi_{s_i}(q^i) + 2\delta b.$$

The second inequality follows from Equation (A.3). For an initially empty graph, $q^i = 0$ and $g(0) = 0$, giving the same bound from initial potential zero. This argument depends on the total inserted mass b, so it also covers mixed edge sizes.

For transport, the old loads satisfy the old capacity and insertion never lowers that capacity. Adding b units can create at most b units of excess. Each successful path reduces excess by its amount (Lemma 4.1); later capacity increases cannot create excess. Total insertion transport is at most b. □

Lemma A.3 (Deletion charge). *A deletion d with reserve destinations satisfies*

$$\delta A_d \leq bZ_d + \Phi_{\text{before}} - \Phi_{\text{after}}, \tag{A.4}$$

where A_d is the transport during that deletion.

The repair is paid for by two sources: the new exposure charge bZ_d and a decrease in stored potential. The bound accounts for both in the same inequality, allowing the potential changes to cancel when we sum over consecutive updates.

Proof of Lemma A.3. Removing the edge decreases every affected load, so it cannot increase potential at the old scale. If the graph becomes empty, there is no transport and the final potential is zero, proving the inequality. Otherwise the raw post-removal vector $\tilde{q}$ satisfies $\tilde{q}_v \leq s_{\text{old}}$ for every vertex.

First consider $s_0 < s_{\text{old}}$. Both endpoints of this scale interval are normalized, so $\beta u \geq 1$ for $s_0 \leq u \leq s_{\text{old}}$. On an active coordinate, Equation (A.3) gives

$$-\frac{\partial}{\partial u}\frac{(q + 1 - \beta u)^2}{u} = \frac{(q+1)^2}{u^2} - \beta^2 \leq \frac{q(q+2)}{u^2}.$$

The last step uses $1/u^2 \leq \beta^2$. A coordinate can activate on this interval only if it belongs to $J_d = \{v : \tilde{q}_v + 1 > \beta s_0\}$. Integrating also accounts for coordinates that are initially inactive:

$$\begin{aligned} \Phi_{s_0}(\tilde{q}) - \Phi_{s_{\text{old}}}(\tilde{q}) &\leq \left(\frac{1}{s_0} - \frac{1}{s_{\text{old}}}\right) \sum_{v \in J_d} \tilde{q}_v(\tilde{q}_v + 2) \\ &\leq \left(1 + \frac{2}{s_{\text{old}}}\right)\left(\frac{s_{\text{old}}}{s_0} - 1\right) \sum_{v \in J_d} \tilde{q}_v \\ &= bZ_d. \end{aligned}$$

The second step uses $\tilde{q}_v + 2 \leq s_{\text{old}} + 2$. If $s_0 \geq s_{\text{old}}$ instead, scale monotonicity gives an increase of at most zero, agreeing with $Z_d = 0$.

After normalization, every pilot-solution replacement or peeling refresh can only increase the scale. Each successful reserve transfer reduces potential by at least δ times its amount (Lemma A.1). Adding these changes to the removal and scale-change bounds proves Equation (A.4). □

A.2.4 Summing the Update Charges. Insertions add only bounded potential, while the exposure charge Z_d bounds the increase caused by lowering the scale after a deletion. These charges pay for all transport in the sequence.

Proof of Lemma 5.4. An insertion transports at most b and increases stable potential by at most $2\delta b$. A deletion transporting A_d units satisfies

$$\delta A_d \leq bZ_d + \Phi_{\text{before}} - \Phi_{\text{after}}.$$

These single-update inequalities are proved in Lemmas A.2 and A.3. Let A_+ and A_- be transport over insertions and deletions, respectively, and Φ_f the final stable potential. Starting from empty gives initial potential zero. Summing the deletion inequalities cancels intermediate potentials except for the increases at insertions, yielding

$$\delta A_- \leq b \sum_d Z_d - \Phi_f + 2\delta b N_+.$$

Since $\Phi_f \geq 0$ and $A_+ \leq bN_+$, we obtain

$$A \leq 3bN_+ + \frac{b}{\delta} \sum_d Z_d.$$

Now use $b = O(r/\epsilon)$ and $1/\delta = (1+\epsilon)/\epsilon = O(1/\epsilon)$. □

A.3 Boundary States and Pilot-Solution Destruction

We now sum the insertion and deletion charges. The intermediate potentials cancel, leaving only the initial and final stable states.

Lemma A.4 (Transport with boundary potentials). *Starting from an empty or normalized certified state with $b \geq \lceil 2r/\epsilon \rceil$ and $0 < \epsilon \leq 1$, let N_+ be the number of insertions and Φ_0, Φ_f the initial and final stable potentials. Set the potential of an empty state to zero. Total transport satisfies*

$$A \leq 3bN_+ + \frac{\Phi_0 - \Phi_f + b\sum_d Z_d}{\delta}.$$

In an r-uniform hypergraph, deleting the sole induced edge of a normalized pilot solution contributes zero exposure.

Proof of Lemma A.4. Let A_+ and A_- be transport summed over insertions and deletions. By Lemma A.2, $A_+ \le bN_+$, and the sum of potential increases over insertions is at most $2\delta bN_+$. The sum of all update-wise potential changes is $\Phi_f - \Phi_0$. Summing Equation (A.4) therefore gives

$$\begin{aligned}\delta A_- &\le b\sum_d Z_d - \sum_d(\Phi_{\text{after}} - \Phi_{\text{before}})\\ &\le b\sum_d Z_d + \Phi_0 - \Phi_f + 2\delta bN_+.\end{aligned}$$

Adding $A_+ \le bN_+$ and dividing the deletion bound by δ proves the stated $3bN_+$ bound. It retains both boundary potentials and applies to every finite prefix of the update sequence.

In an r-uniform hypergraph, a one-edge pilot solution has size $k \ge r$, while normalization gives $1/k \ge 1/r$. Hence $k = r$ and its old density is exactly $1/r$. Any surviving edge supplies at least this density, so replacing a destroyed pilot solution cannot lower the scale. An empty result has zero exposure by definition. □

Exposure is confined to a degree tail because $\tilde{q}_v \le b\deg_{\text{after}}(v)$. For destruction in mixed ranks, if the old pilot solution has size k and the fallback edge has size f, then $s_{\text{old}}/s_0 \le f/k$; in particular, $f \le k$ gives zero exposure. Thus the charge depends on the exposed degree tail and the drop in pilot-solution density. Combining its transport bound with Equation (8) bounds the associated search work. The new charge need not be smaller on every destructive deletion: one may replace Z_d there by $\min\{Z_d, \delta(1+\delta)m_{\text{after}}\}$, using the old excess and stable-potential bounds before telescoping. Indeed, after removal there are bm_{after} units in total and capacity only increases during repair, so $A_d \le bm_{\text{after}}$. The final state satisfies $\Phi_{\text{after}} \le \delta^2 bm_{\text{after}}$, while $\Phi_{\text{before}} \ge 0$. Consequently

$$\delta A_d + \Phi_{\text{after}} - \Phi_{\text{before}} \le \delta(1+\delta)bm_{\text{after}}.$$

Both this inequality and the bZ_d bound concern the same deletion and the same endpoint potentials, so their right-hand sides may be minimized before summing. An empty result has $m_{\text{after}} = 0$.

A.4 A Degree-Tail Specialization

The exposure bound gives a more explicit guarantee when few vertices have degree near the repair threshold. Consider an internal deletion whose old pilot solution has $M \ge 2$ edges and k vertices. Set $M' = M - 1$, $s = s_{\text{old}} = \alpha bM/k$, and $s' = \alpha bM'/k$. Let

$$H_d = \{v : b\deg_{\text{after}}(v) > \beta s' - 1\}.$$

Normalization ensures $s_0 \ge s'$, so every coordinate contributing to Z_d lies in H_d. Moreover, $s/s_0 - 1 \le 1/M'$ and $\tilde{q}_v \le s$. Hence

$$Z_d \le \frac{s+2}{bM'}|H_d| \le 2(1+2\epsilon)\frac{|H_d|}{k}. \tag{A.5}$$

The last step uses $M/M' \le 2$ and $1/s \le \delta/2$.

Corollary A.5 (Uniform hypergraphs with a small degree tail). *Consider an r-uniform update sequence satisfying the assumptions of Lemma A.4. If $|H_d| \le \eta|C_d|$ for every internal deletion that leaves an edge in its old pilot solution, then*

$$A = O\Big(bN_+ + \frac{\Phi_0}{\epsilon} + \frac{b\eta}{\epsilon}N_-\Big),$$

where N_- is the number of deletions and C_d is the pre-deletion pilot solution. In particular, the result applies to ordinary graphs ($r = 2$).

For an empty start and $b = \Theta(r/\epsilon)$, the bound is $O((r/\epsilon)N_+ + (r\eta/\epsilon^2)N_-)$ units of transport. The parameter η measures the size of the relevant degree tail relative to the pilot solution, so this conclusion is useful when that ratio stays small throughout the sequence.

Proof. External deletions leave the pilot-solution density unchanged. In a uniform hypergraph, destruction of a normalized one-edge pilot solution has zero exposure by Lemma A.4. For every remaining deletion, Equation (A.5) gives $Z_d = O(\eta)$. Substitute their sum in Lemma A.4, drop $\Phi_f \ge 0$, and use $1/\delta = (1+\epsilon)/\epsilon = O(1/\epsilon)$. □

For mixed ranks, the same bound retains the additional term $(b/\delta)\sum_{d:M_d=1} Z_d$ for pilot-solution destruction. The degree-tail condition permits many low-degree vertices; it concerns the degree threshold at each deletion and need not follow from a fitted power-law exponent. This is a transport bound; search and endpoint costs are incorporated in Section 5.3.

B SEARCH COUNTS AND COMPLETE TIME ACCOUNTING

We first bound the number of searches over an update sequence, then add the costs of normalization, heap maintenance, and occasional peeling. Throughout this subsection, h bounds the high-load region at every search and $\mathcal{J} = I$ denotes the active incidence volume.

Making the search count explicit. Start from an empty state or a normalized certified state, and let $\bar{\rho} \ge 1/r$ bound the optimum over all states, including the initial state. Before an update, stability gives $q_v \le \alpha b\bar{\rho}$. An insertion adds only b units, while a deletion decreases loads. During repair, a transfer decreases the source load, leaves intermediate loads unchanged, and keeps its destination at most $c \le \alpha b\bar{\rho}$. Thus every search has $Q \le \alpha b\bar{\rho} + b$. For an internal deletion let M_d, k_d be the old pilot solution's edge count and size. Define

$$V_a = \sum_{M_d \ge 2}\left\lceil\frac{ab}{k_d}\right\rceil + \sum_{M_d=1}\left(\left\lfloor\frac{ab}{k_d}\right\rfloor - \left\lfloor\frac{ab}{r}\right\rfloor\right).$$

The quantity V_a bounds downward changes of the normalized integer target, assigning an empty state the artificial baseline $\lfloor ab/r\rfloor$. For nonempty states this target is $\lfloor abL\rfloor$. The first sum covers a pilot solution that loses one of several edges; the second covers loss of its sole edge and subsequent normalization or an empty state. For transport we need only decreases before nonempty repairs, since an emptying update requires no transport. We use V_α for those outer capacity decreases and V_γ for the destination target.

Lemma B.1 (Transport and failed-search counts). *For* CAP, *if the high-load region has at most h vertices at every search, then*

$$\begin{aligned}A &\le bN_+ + hV_\alpha,\\ F &\le \lfloor\gamma b\bar{\rho}\rfloor - \lfloor\gamma b/r\rfloor + V_\gamma.\end{aligned}$$

The first bound charges successful transport to inserted mass and capacity decreases. The second charges failed searches to upward threshold progress: each failure increases the threshold, and

only deletions can make that progress need to be repeated. Together they supply search counts for Equation (8).

Proof of Lemma B.1. By Lemma A.2, insertion transport is at most b. For a deletion requiring repair, let c_{old} be its old capacity and c_{start} the capacity at its first search, after normalization and any initial peeling. Raw removal only decreases loads, so every load is at most c_{old}. At this first search, every overloaded vertex belongs to its high-load region, which contains at most h vertices. The initial excess is therefore at most $h(c_{\text{old}}-c_{\text{start}})_+$. During repair, capacities only increase and each successful path reduces excess by its amount. Thus this initial excess bounds total deletion transport.

If $M_d \geq 2$, the old pilot solution retains density $(M_d-1)/k_d$ before normalization. Subsequent normalization can only improve it. For any $a > 0$, the threshold decrease is consequently at most

$$\left\lfloor \frac{abM_d}{k_d} \right\rfloor - \left\lfloor \frac{ab(M_d-1)}{k_d} \right\rfloor \leq \left\lceil \frac{ab}{k_d} \right\rceil.$$

This follows from $\lfloor x \rfloor - \lfloor x-y \rfloor \leq \lceil y \rceil$. If $M_d = 1$ and an edge survives, normalization gives density at least $1/r$, so the decrease is at most $\lfloor ab/k_d \rfloor - \lfloor ab/r \rfloor$. This quantity is nonnegative since the old normalized pilot solution has $k_d \leq r$. External deletions do not decrease the pilot-solution density, and deletion of the last active edge requires no transport. Summing with $a = \alpha$ bounds A.

For failed searches, define $t_* = \lfloor \gamma b/r \rfloor$ and track $\tau = t$ in nonempty states and $\tau = t_*$ in empty states. Normalization always gives $t \geq t_*$. Every failure increases this integer target by at least one (Equation (6)). Insertions and density-improving pilot-solution replacements cannot decrease it. The same floor-difference calculation with $a = \gamma$ bounds its total downward variation by V_γ. In particular, on an emptying update the previous state has one active edge, and its target drops by at most $\lfloor \gamma b/k_d \rfloor - t_*$; the next insertion cannot take it below t_*. Thus empty states introduce no additional charge. Since $t_* \leq \tau \leq \lfloor \gamma b \bar{\rho} \rfloor$, summing target increases and decreases gives the claimed bound on F. □

Lemma A.4 gives an alternative bound on A for reserve repair; the smaller bound can be substituted into Equation (8). A fixed-size high-degree core with all other vertices below the actual repair threshold is sufficient. Degrees include crossing edges. Snapshot density separation alone is insufficient.

Fallback and maintenance costs. Failed searches and peeling supply known induced-edge counts. For an unknown-count fallback S, a separate edge stamp deduplicates encounters in its positive incidence lists before testing containment. The test stops at the first endpoint outside S. With analogous counters P_S, E_S, J_S,

$$P_S \leq \sum_{v \in S} q_v, \qquad E_S \leq P_S, \qquad J_S \leq \min\{r, |S|+1\} E_S.$$

Every internal edge has a positive owner in S and is counted once. The fallback edge itself is fully tested, so $J_S \geq |S|$ also pays for copying and marking the new pilot solution. We retain $Z_{\text{norm}} = \sum_S (P_S + J_S)$, as mixed-rank fallback need not cost $O(b)$.

Let $R_{\text{up}} = \sum_{\text{updates}} |e|$, K count successful searches, $\Lambda = \log(2 + n + \mathcal{J}_{\max})$, D_0 be inherited refresh debt, and Π_0 be initial container-growth credit. For each successful path, h_{path} is the size of H at its search. With compact or preallocated edge IDs, the single-path implementation has the following detailed accounting. Its first line pays for inherited state credits and edge updates, its second for search inspections, fallbacks, and heap changes, its third for locating receiving shares, and its last for peeling:

$$\begin{aligned} O\big(&\Pi_0 + |C_0| + R_{\text{up}}[\log(r+1) + \log(b+1) + \log(n+1)] \\ &+ P_{\text{tot}} + J_{\text{tot}} + Z_{\text{norm}} + K\log(n+1) \\ &+ \sum_{\text{paths}} [(\ell-1)\log(h_{\text{path}}+1) + \log(r+1)] \\ &+ (P_{\text{tot}} + D_0)\Lambda\big). \end{aligned} \tag{B.1}$$

We justify each term in Equation (B.1) below.

Edge updates and path transfers. For an updated edge of size s_e, sorting and locating endpoint slots cost $O(s_e \log(r+1))$. Discrete water filling searches at most $b+1$ integer levels, scanning the s_e endpoints in each binary-search step, and costs $O(s_e \log(b+1))$. Heap changes contribute $O(s_e \log(n+1))$. Summing s_e gives the R_{up} term. Positive-incidence lists use inverse positions, so changing one endpoint share costs amortized constant time apart from its slot lookup. The initial container-growth credit Π_0 pays any inherited allocation cost not charged to updates in the sequence.

Search exploration contributes $P_{\text{tot}} + J_{\text{tot}}$. Its parent pointers already identify donor slots. A receiving slot is found by binary search in the sorted endpoints. As in the proof of Lemma 5.3, every nonterminal path edge was fully expanded before a destination was found, so all of its endpoints lie in the high-load region. Only the terminal edge can have rank exceeding h_{path}, giving the displayed sum of lookup costs. Intermediate load changes cancel, so a successful path changes only two heap keys, contributing $O(\log(n+1))$ per success. Path traversal is charged to its discovering search.

Pilot-solution installation and budgeted peeling. A failed search already supplies its reached vertices and induced-edge count. A fallback costs $P_S + J_S$, and a peeling pass supplies its new pilot solution as part of that pass. The total cost of clearing old membership marks is at most the initial $|C_0|$ plus the total sizes of subsequently installed pilot solutions. These sizes are charged to the corresponding searches, fallbacks, or peeling passes.

At a peeling refresh j, let D_j be the accumulated positive-scan debt, and let $m_j, \mathcal{J}_j$ describe the current graph. The trigger requires $D_j > 4(n + rm_j)$. Since $\mathcal{J}_j \leq rm_j$, a pass costs

$$O((n + \mathcal{J}_j)\log(2 + n + \mathcal{J}_j)) = O(D_j \Lambda).$$

Debt is reset after each pass, so the charged scan intervals are disjoint and $\sum_j D_j \leq P_{\text{tot}} + D_0$. This proves the final term of Equation (B.1), even if the graph size changes between refreshes. It also includes constructing and marking the new peeling pilot solution.

Construction, reporting, and space. An empty state takes $\Theta(n)$ time to construct. For a nonempty initial graph, we also charge the work of inserting its initial edges. Query materialization adds $\Theta(1 + |C|)$ per reported set. The bounds use representable unit-cost integer arithmetic and compact or preallocated edge identifiers, with allocation charged to the update sequence. A compacted state occupies $O(n + I)$ words. The prototype retains deleted edge vectors and allocated container capacities, so its storage also depends on the update history. Independent correctness checking is performed outside the timed updates.

Deriving the summary guarantees.

Proof of Theorems 5.1 and 5.2. Starting empty gives $C_0 = \emptyset$, $D_0 = 0$, and no inherited container-growth credit; initialization costs $O(n)$. Write $S_{\text{scan}} = P_{\text{tot}} + J_{\text{tot}}$. Every successful search inspects an incidence, so $K \le S_{\text{scan}}$. The incidences along a returned path have already been examined by that search, so the total number of path steps is also at most S_{scan}. The nonempty update sequence gives $m_* \ge 1$. Since $I_* \le rm_*$ and $b = \lceil 2r/\epsilon \rceil$, all logarithms in Equation (B.1) are $O(\log((n+rm_*)/\epsilon))$. Substitution yields Equation (7), including the cost of peeling.

For the coarse bound, a search inspects at most I_* positive incidences and at most I_* endpoints: each vertex is expanded once and each edge is stamped after its first expansion. Its inspection cost is at most $2I_*$. An insertion transports at most b units. During a deletion, the initial excess is at most the remaining total mass bm_*; capacities only increase during repair, so deletion transport is at most bm_*. Integer augmentation therefore bounds successful searches per update by bm_*.

Each failed search increases the inner threshold by a factor of at least $1 + \epsilon/4$ (Equation (6)). The initial threshold is at least $\lfloor \gamma b/r \rfloor$ and the final one is at most $\lfloor \gamma bm_* \rfloor$. Hence there are $O(\epsilon^{-1} \log(2 + rm_*))$ failures per update. Consequently,

$$S_{\text{scan}} = O\big(NI_* \left[bm_* + \epsilon^{-1} \log(2 + rm_*)\right]\big).$$

There is at most one fallback normalization per update. Its stamped incidence and endpoint scans each cost at most I_*, giving $Z_{\text{norm}} = O(NI_*)$, while $R_{\text{up}} \le Nr$. Finally, $b = O(r/\epsilon)$ and $\log(2 + rm_*) = O(rm_*)$. Substitution in Equation (7) gives

$$T_{\text{maint}} = O\left(n + N\frac{rm_*I_*}{\epsilon} \log \frac{n + rm_*}{\epsilon}\right). \tag{B.2}$$

Excluding initialization and dividing by N proves Theorem 5.1. The approximation follows from Theorem 4.3; the density and pilot solution are stored explicitly, giving the stated query costs. □

The main text proves Lemma 5.6 by substituting the locality and transport bounds into Equation (7).

C CQ RECONSTRUCTION AND EVALUATION PROTOCOL

Our `CQ` baseline reconstructs the basic amortized graph core, nested label buckets, and shared-copy representation from the earlier Chekuri–Quanrud preprint whose results were incorporated into Chekuri et al. [1], using the preprint's Section 6 hypergraph extension. That preprint is linked in Section 6.1. Each logical update executes K unit-copy operations; copies with the same hyperedge and head share a multiplicity and endpoint labels. There are no bulk transfers or empirically tuned copy counts.

Fixed parameters and extraction. Let n be the size of the fixed vertex universe and r the maximum hyperedge rank over the complete update sequence. For requested $0 < \epsilon \le 1$, we set

$$q = (1+\epsilon)^{1/4}, \qquad M = \left\lceil \frac{\log \max\{2, n\}}{\log q} \right\rceil,$$
$$a = \exp\left(\frac{\log(1+\epsilon)}{8M}\right) - 1, \qquad P = (1+a)^2,$$
$$\beta = 3 + 3a, \qquad \Delta = (1+\epsilon)^{-1/2} - (1+\epsilon)^{-1},$$

and $K = \lceil \beta Mr/(q\Delta) \rceil + 1$. These explicit constants are our conservative choice, not constants specified in the source paper. At quiescence, the label invariants imply $d(h) \le P\, d(u) + 3 + 2a \le P\, d(u) + \beta$ for each copy head h and every endpoint u. With $\mu = \max_v d(v)$, start at $t_0 = \mu$ and use $t_{i+1} = (t_i - \beta)/P$ and $S_i = \{v : d(v) \ge t_i\}$, including all threshold ties. We return S_{i+1} at the first index with $|S_{i+1}| \le q|S_i|$ and compute its fully induced logical-edge density. For a nonempty snapshot satisfying these invariants,

$$\rho(S_{i+1}) \ge \frac{\rho^*}{qP^M} - \frac{\beta M}{qK} \ge \frac{\rho^*}{1+\epsilon},$$

using $\rho^* \ge 1/r$; empty snapshots return the empty set.

Deletion completion and implementation scope. Section 6 of that preprint does not specify hypergraph deletion pseudocode. Our completion selects a global minimum-load endpoint for a unit reorientation, relabels the destination group, and schedules both changed endpoints. A triggering endpoint whose condition remains violated stays on the worklist; residual source-group copies retain their labels. We have not established general termination or the original amortized update-time bound for this worklist. We do not implement the deamortized or full worst-case construction. Direct group-specific endpoint labels and links require $O(\sum_{e \in E} |e|^2)$ worst-case live storage, in addition to the vertex state and registry capacity determined by peak simultaneous groups. Retired group identifiers are reused. Extraction scans at most M thresholds and the active edge incidences, taking $O(Mn + p)$ time for $p = \sum_{e \in E} |e|$. We do not claim its original time or space bounds.

Result admission. Each reported `CQ` mean requires three complete repetitions, with the entire update sequence and every scheduled query processed. Independent replay checks the returned sets' induced densities against exact snapshot optima and the requested rational accuracy. Passing prefixes and single decision trials do not constitute three-run results. The 1,800-second `CQ` wall limit includes input loading, invariant checks, and allocation output; these are excluded from running time. We sample working-set and private-memory use against a 24 GiB limit rather than impose a hard operating-system cap. Running-time means follow the reporting criteria in Section 6.1.

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